
Homework 1

Michael Introini

10 July 2017

Chapter 0

Problem 5

If C is a set with c elements, how many elements are in the power set of C ? Explain your answer.

There are $2^{|C|}$ elements in the power set of C . This is because the power set C is the set of every possible subset of C .

Problem 7

For each part, give a relation that satisfies the condition.

a. Reflexive and symmetric but not transitive

$$S = \{1, 2, 3\}$$

$$R = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (3, 1), (3, 2), (3, 3)\}$$

b. Reflexive and transitive but not symmetric

$$S = \{1, 2, 3\}$$

$$R = \{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$$

c. Symmetric and transitive but not reflexive

$$S = \{1, 2, 3\}$$

$$R = \{(1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2)\}$$

Problem 10

Find the error in the following proof that $2 = 1$. Consider the equation $a = b$. Multiply both sides by a to obtain $a^2 = ab$. Subtract b^2 from both sides to get $a^2 - b^2 = ab - b^2$. Now factor each side, $(a + b)(a - b) = b(a - b)$, and divide each side by $(a - b)$ to get $a + b = b$. Finally, let a and b equal 1, which shows that $2 = 1$.

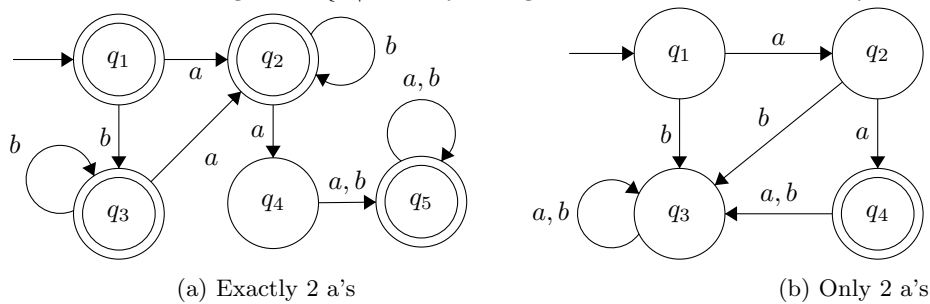
The error in this proof lies in the fact that letting $a = 1, b = 1$ needs to hold true for every part of the proof. For example, $a = 1, b = 1$ holds true for the first 3 steps of the proof. However, when stating that one should divide to obtain $a + b = b$, you would actually be dividing by zero, which is undefined. Thus contradicting the original proof.

Chapter 1

Problem 1.5g

Each of the following languages is the complement of a simpler language. In each part, construct a DFA for the simpler language, then use it to give the state diagram of a DFA for the language given. In all parts, $\Sigma = \{a, b\}$.

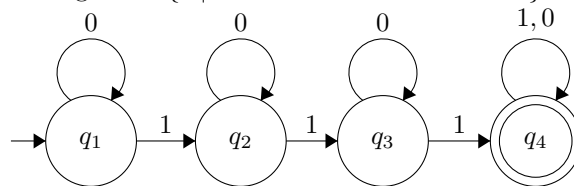
Figure 1: $\{w \mid w \text{ is any string that doesn't contain exactly two } a's\}$



Problem 1.6b

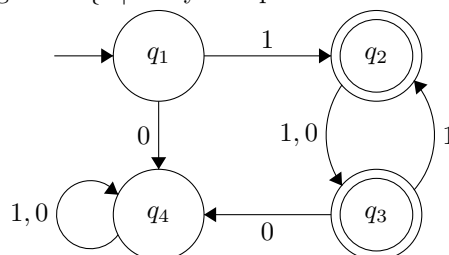
Give state diagrams of DFAs recognizing the following languages. In all parts, the alphabet is $\{0, 1\}$.

Figure 3: $\{w \mid w \text{ contains at least three } 1s\}$



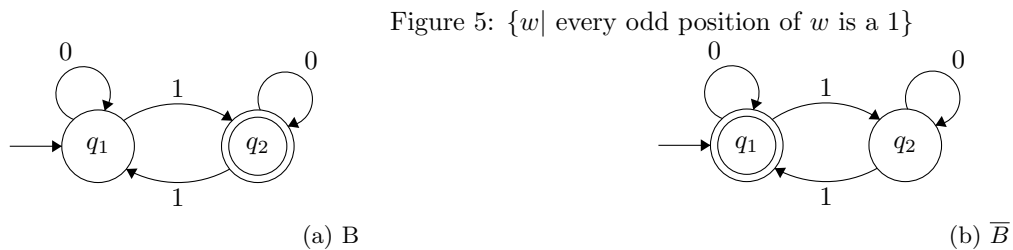
Problem 1.6i

Figure 4: $\{w \mid \text{every odd position of } w \text{ is a } 1\}$



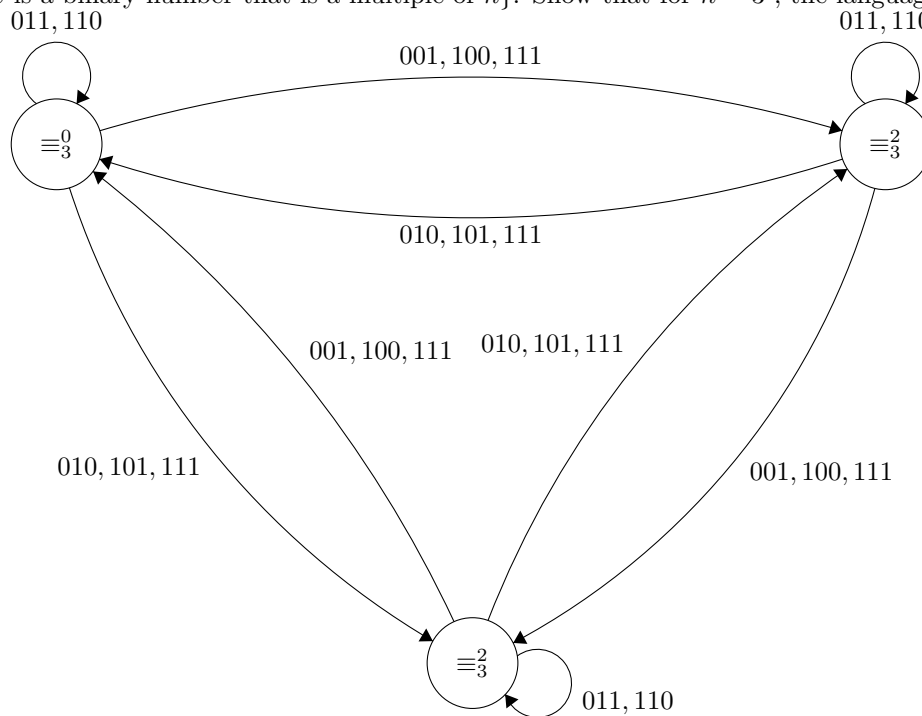
Problem 1.14a

Show that if M is a DFA that recognizes language B , swapping the accept and nonaccept states in M yields a new DFA recognizing the complement of B . Conclude that the class of regular languages is closed under complement.



Problem 1.37

Let $C_n = \{x \mid x \text{ is a binary number that is a multiple of } n\}$. Show that for $n = 3$, the language C_n is regular.



Problem 1.48

Let $\Sigma = \{0, 1\}$ and let

$D = \{w \mid w \text{ contains an equal number of occurrences of the substrings } 01 \text{ and } 10\}$.

Thus $101 \in D$ because 101 contains a single 01 and a single 10 , but $1010 \notin D$ because 1010 contains two 10 s and one 01 . Show that D is a regular language.

